SUBJECT GUIDE

Quantitative aptitude & maths

Every quant topic of SSC, Banking and Railway mapped to the NCERT chapter that teaches it.

8	23	34
Modules	Topics	NCERT chapters

Asked in: SSC · Banking · Railway · Insurance · Police

Contents

1 Number system
2 Percentage, profit & loss, interest
3 Ratio, proportion, time & work, speed
4 Algebra
5 Geometry & coordinate geometry
6 Mensuration
7 Trigonometry
8 Statistics, probability & data interpretation

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MODULE 1

Number system

- ☐ Integers, rational & real numbers
- ☐ HCF & LCM, divisibility
- ☐ Squares, cubes, powers, surds

How it's asked: Remainder and unit-digit questions take seconds once rules are learnt.

READ (FREE, OFFICIAL)

NCERT Class 10 · Real Numbers · PDF ↗
NCERT Class 8 · Rational Numbers · PDF ↗
NCERT Class 8 · Squares and Square Roots · PDF ↗
NCERT Class 8 · Cubes and Cube Roots · PDF ↗
NCERT Class 8 · Exponents and Powers · PDF ↗
NCERT Class 9 · The World of Numbers · PDF ↗

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MODULE 2

Percentage, profit & loss, interest

- ☐ Percentages
- ☐ Profit, loss, discount
- ☐ Simple & compound interest

How it's asked: Highest-weight arithmetic in SSC and Banking prelims.

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NCERT Class 8 · Comparing Quantities · PDF ↗

NCERT Class 7 · Comparing Quantities · PDF ↗

3

MODULE 3

Ratio, proportion, time & work, speed

- ☐ Ratio & proportion, partnership, mixtures
- ☐ Time & work, pipes
- ☐ Speed, distance, time; trains & boats

How it's asked: Use the LCM method for time & work; relative speed for trains.

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NCERT Class 8 · Direct and Inverse Proportions · PDF ↗

NCERT Class 8 · Proportional Reasoning-1 · PDF ↗

NCERT Class 8 · Proportional Reasoning-2 · PDF ↗

4

MODULE 4

Algebra

- ☐ Polynomials & identities
- ☐ Linear & quadratic equations
- ☐ Arithmetic progressions

How it's asked: SSC CGL: identities like a^3+b^3 and x + 1/x types.

READ (FREE, OFFICIAL)

NCERT Class 10 · Polynomials · PDF ↗

NCERT Class 10 · Pair of Linear Equations in Two Variables · PDF ↗

NCERT Class 10 · Quadratic Equations · PDF ↗

NCERT Class 10 · Arithmetic Progressions · PDF ↗

NCERT Class 8 · Algebraic Expressions and Identities · PDF ↗

NCERT Class 9 · Exploring Algebraic Identities · PDF ↗

5

MODULE 5

Geometry & coordinate geometry

- ☐ Lines & angles, triangles, quadrilaterals, circles
- ☐ Similarity & congruence
- ☐ Coordinate geometry

How it's asked: Triangle centres and circle theorems are SSC favourites.

READ (FREE, OFFICIAL)

NCERT Class 10 · Triangles · PDF ↗
NCERT Class 10 · Coordinate Geometry · PDF ↗
NCERT Class 10 · Circles · PDF ↗
NCERT Class 8 · Understanding Quadrilaterals · PDF ↗
NCERT Class 9 · Orienting Yourself: The Use of Coordinates · PDF ↗

6

MODULE 6

Mensuration

- ☐ Area & perimeter
- ☐ Surface area & volume of solids
- ☐ Areas related to circles

How it's asked: Keep a one-page formula sheet for 2D and 3D shapes.

READ (FREE, OFFICIAL)

NCERT Class 10 · Areas Related to Circles · PDF ↗
NCERT Class 10 · Surface Areas and Volumes · PDF ↗
NCERT Class 8 · Mensuration · PDF ↗
NCERT Class 9 · Measuring Space: Perimeter and Area · PDF ↗

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MODULE 7

Trigonometry

- ☐ Ratios & identities
- ☐ Heights & distances

How it's asked: Standard-angle values and identities answer most questions.

READ (FREE, OFFICIAL)

NCERT Class 10 · Introduction to Trigonometry · PDF ↗
NCERT Class 10 · Some Applications of Trigonometry · PDF ↗

8

MODULE 8

Statistics, probability & data interpretation

- ☐ Mean, median, mode
- ☐ Probability
- ☐ Tables, bar & pie charts (DI)

How it's asked: Banking mains DI: approximate quickly; don't calculate exactly.

READ (FREE, OFFICIAL)

NCERT Class 10 · Statistics · PDF ↗

NCERT Class 10 · Probability · PDF ↗

NCERT Class 8 · Data Handling · PDF ↗

NCERT Class 9 · The Mathematics of Maybe: Introduction to Probability · PDF ↗

NCERT Class 11 · Organisation of Data · PDF ↗

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